Final Year Research Project Presentation

Analyzing the Information Loss in India’s AQI: A Multi-City, Multi-Profile Audit Using Total Respiratory Burden

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Outline

Air Quality Crisis in India

The national scale

- 14/15 of world's most polluted cities are in India
- 1.6M deaths/year $\rightarrow$ one every 20 seconds
- Average Indian loses 5.3 years of life expectancy

Delhi as the extreme case

- Mean PM2.5: 152 μg/m^3 $\rightarrow$ 30$\times$ WHO (5 μg/m^3)
- Mean PM10: 290 μg/m^3 $\rightarrow$ 19.3$\times$ WHO (15 μg/m^3)
- November PM2.5 exceeds 300 μg/m^3 $\rightarrow$ schools close, hospital admissions surge

The public's only window

- The Air Quality Index (AQI): one number, six colour codes
- AQI 101 and 199 share the same "Moderate" label
- But actual lung dose triples from bottom to top of that range

Core Question: How much physiologically relevant information does AQI actually retain?

AQI as a Lossy Compression System

Indian AQI reduces $PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, O_3, NH_3 to one value:

0. Temporal averaging:

$$\bar{C}_p = \text{(t - 24) hr rolling mean of pollutant } p$$

1. Sub-index normalisation (forced common scale):

$$\bar{C}_p \longrightarrow I_p \in [0, 500]$$

2. Max-pooling (pick worst, discard rest):

$$\text{AQI} = \max\left(I_{\text{PM}_{2.5}},\ I_{\text{PM}_{10}},\ I_{\text{SO}_2},\ I_{\text{NO}_2},\ I_{\text{CO}},\ I_{\text{O}_3},\ I_{\text{NH}_3}\right)$$

3. Categorization (6 fixed bands):

Good	Satisfactory	Moderate	Poor	Very Poor	Severe
0–50	51–100	101–200	201–300	301–400	401–500

What We Did – A Mechanistic Audit

- Used Total Respiratory Burden (TRB) – a physiologically grounded metric; comes from:
  1. ICRP deposition model
  2. Gas uptake models.
- Analysed **56,880 hours** of CPCB data from Delhi, Mumbai, Hyderabad (Jan 2024 – Feb 2026).
- Audited the AQI using three complementary lenses:
  1. **Information theory** – how much mutual information survives?
  2. **Within-bucket analysis** – how much dose vary inside a single category?
  3. **Change-point detection** – do official boundaries match data-driven thresholds?

Prior Work: Limitations of Single-Pollutant AQIs

Study	Key Finding
Cromar et al. (2020)	EPA AQI under-reports multi-pollutant risk. Additive index better matches ER visits.
George et al. (2025)	City-specific AQHI improves mortality prediction by 18–22% over NAQI.
Tan et al. (2020)	Multi-pollutant indices beat max-pooling in 71% of studies; 15–30% stronger correlation with respiratory outcomes.
Kladakis et al. (2025)	PM10+O3+NO2 together amplify hospital admissions by factor 2.4 vs. single pollutants.
Hu et al. (2015)	Max-pooling AQI misclassified 23% of high-risk days as moderate/good in China.

Critical Gap: no formal information-theoretic audit of India's AQI using a physiologically grounded dose metric.

Data Pipeline: From Raw CPCB to Clean Dataset

Source: CPCB CAAQMS portal
Cities: Delhi, Mumbai, Hyderabad
Period: 1 Jan 2024 – 28 Feb 2026 (26 months)
Resolution: Hourly
Raw size: 56,880 rows $\times$ 9 cols

Cleaning steps:

1. Unit standardization: mg/m^3 $\to$ μg/m^3
2. Datetime conversion: parsed to `datetime64[ns]`.
3. Missing imputation: grouped by city, sorted chronologically, forward-fill then backward-fill (<5% missing)
4. Physical constraint: PM2.5 capped to PM10 (557 violating rows fixed).

Output: 56,880 rows, zero missing values, no negatives, no PM2.5 > PM10 violations.

Total Respiratory Burden (TRB)

TRB = total mass of pollutants deposited/absorbed in the human respiratory tract per hour.

$$D_{i,h,p} = C_{i,h} \times V_{E,p} \times 60 \times F_{\text{total},i} \quad [\mu\text{g/hour}]$$

$$\text{TRB}_{\text{actual},h,p} = \sum_{i=1}^{7} D_{i,h,p}$$

- $C_{i,h}$: ambient concentration (µg m^{-3}) from CPCB.
- $V_{E,p}$: minute ventilation (m^3/min) – ICRP values (4 profiles).
- $F_{\text{total},i}$: total deposition/uptake fraction – ICRP for PM2.5/PM10; gas models for others.

TRB reflects real-time multi-pollutant co-exposure.

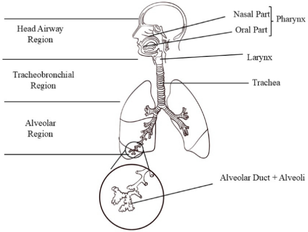
Demographic Profiles and Gas Uptake Fractions

ICRP minute ventilation V_E

Profile	V_E (m^3/min)	Code
Male Sitting	0.0090	m_s
Male Walking	0.0250	m_w
Female Sitting	0.0084	f_s
Female Walking	0.0210	f_w

Gas uptake fractions (mid scenario)

Pollutant	HA	TB	AL	Total UF	F_{total}
CO	0.00	0.00	0.90–0.98	0.90–0.98	0.94
SO_2	0.75–0.85	0.02–0.10	0.00	0.80–0.90	0.85
NH_3	0.75–0.85	0.00–0.05	0.00–0.05	0.80–0.90	0.85
O_3	0.05–0.20	0.30–0.60	0.20–0.50	0.70–0.90	0.80
NO_2	0.05–0.20	0.25–0.45	0.20–0.65	0.50–0.80	0.65



Adapted from Singh et al.(2025)

HA = Head Airways
TB = Tracheobronchial
AL = Alveolar

EDA Highlights

- **Delhi extreme seasonal shift:** Nov mean AQI 568 vs. July 144 ($3.95\times$); Oct→Nov +140 – sharpest regime shift.
- **WHO vs. NAAQS:** Delhi exceeds WHO in 80.6% of hours, NAAQS only 28.0% – regulatory standard less protective.
- **TRB uncertainty:** Relative spread (low/high gas-uptake) = 15–16% across all demographics.
- **Sub-index gap (low, <20):** Delhi 20.1% vs. Hyderabad 59.4% of hours — competitive conditions drive steeper dose gradients in Hyderabad.

Information-Theoretic Estimation

Mutual Information $I(X;Y)$ (nats) measures how much knowing X reduces uncertainty about Y. Captures linear and nonlinear associations.

KSG estimator (Kraskov et al., Phys. Rev. E, 2004):

- Type II – corrects for asymmetry i.e. 7D (input) vs. 1D (target).
- $k = 4$ neighbours, added 10^{-10} Gaussian noise.

Quantities computed per city:

1. $I(P_7; \text{TRB})$ – joint MI of all 7 pollutants
2. $I(\text{AQI}_{\text{cont}}; \text{TRB})$ and $I(\text{AQI}_{\text{cat}}; \text{TRB})$
3. Compression ratios r_{cont}, r_{cat}

Quantifying information loss:

- Absolute loss (nats): $I(P_7; \text{TRB}) - I(\text{AQI}; \text{TRB})$
- Relative loss (%): $(1 - r) \times 100$

Information Upper Bound & Absolute Loss

Abs. loss = $I(P_7; \text{TRB}) - I(\text{AQI}; \text{TRB})$ – information discarded by AQI.

City	$I(P_7; \text{TRB})$ (nats)	$I(\text{AQI}_{\text{cont}}; \text{TRB})$ (nats)	Abs. loss (cont)(nats)	$I(\text{AQI}_{\text{cat}}; \text{TRB})$ (nats)	Abs. loss (cat)(nats)
Delhi	1.87	0.32	1.55	0.261	1.609
Mumbai	1.60	0.37	1.23	0.216	1.384
Hyderabad	2.02	0.12	1.90	0.073	1.947

- Raw P_7 vector carries **1.23–1.90 nats more** dose info than continuous AQI.
- Variation across 12 TRB targets (4 profiles $\times$ 3 gas scenarios) is < 0.003 nats – upper bound is property of raw signal, not demographics.

Compression Ratios: How Much Information Survives?

$$\text{Compression ratio:} \qquad r = \frac{I(\text{AQI}; \text{TRB})}{I(P_7; \text{TRB})}$$

AQI compression ratios (mid scenario, averaged over four profiles).

City	Continuous AQI			Categorical AQI		
	$I(\text{AQI}_{\text{cont}}; \text{TRB})$ (nats)	r_{cont}	Loss (%)	$I(\text{AQI}_{\text{cat}}; \text{TRB})$ (nats)	r_{cat}	Loss (%)
Delhi	0.320	0.171	82.9	0.261	0.140	86.0
Mumbai	0.372	0.232	76.8	0.216	0.134	86.6
Hyderabad	0.124	0.062	93.8	0.073	0.037	96.3

Bootstrap 95% CI width = 0.004 ratio units – estimates are precise and stable.

Interaction Information: Synergy Lost by Max-Pooling

Interaction information (II): What do two pollutants tell us together vs. separately?

$$II(p_i, p_j; \text{TRB}) = I(p_i, p_j; \text{TRB}) - \big(I(p_i; \text{TRB}) + I(p_j; \text{TRB})\big)$$

- $II > 0 \rightarrow$ **synergy**: together they reveal more than the sum of their individual info
- $II < 0 \rightarrow$ **redundancy**: they mostly say the same thing

Pollutant pair	Delhi (nats)	Mumbai (nats)	Hyderabad (nats)
PM10 $\times$ CO	**+0.443** (syn)	**+0.374** (syn)	**+0.351** (syn)
PM2.5 $\times$ CO	+0.140 (syn)	+0.196 (syn)	+0.154 (syn)
PM2.5 $\times$ PM10	-0.346 (redn)	-0.231 (redn)	-0.066 (redn)
CO $\times$ O3	+0.038 (syn)	**+0.151** (syn)	+0.023 (syn)

- **PM10$\times$CO** = largest synergy in every city — the exact synergy lost when AQI picks PM10 and discards CO.
- Mumbai's **CO$\times$O3** synergy (+0.151) reflects coastal photochemistry.

CO: The AQI's Informational Blind Spot

City	$I(\mathrm{CO};\mathrm{TRB})$ (nats)	Ratio to 2nd pollutant	CO responsible for AQI
Delhi	0.88	2.9× (PM10)	2.86% of hours
Mumbai	0.83	3.1× (NO2)	4.25% of hours
Hyderabad	1.17	3.8× (PM10)	7.58% of hours

Why CO dominates (and is harmful)?

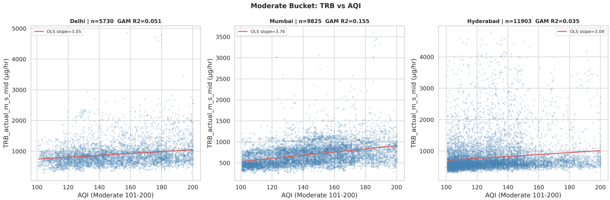
- Strongly correlates with combustion activity (traffic, industry, biomass).
- Near-complete AL deposition **(0.90–0.98)** & highest Total uptake fraction**(0.94)**.
- Forms **carboxyhaemoglobin** – reduces blood oxygen-carrying capacity.
- $230\times$ O_2 affinity for blood | CVD ↑ with CO ↑ | ∼29k deaths/yr

Within-Bucket Gradient: Inside Category, Dose Can 3x

- The AQI category Moderate (101–200) is the most common in all three cities:
  - Delhi: 30%, Mumbai: 52%, Hyderabad: 63% of all hours
- *How much actual lung dose (TRB) change as AQI moves from 101 to 200?*

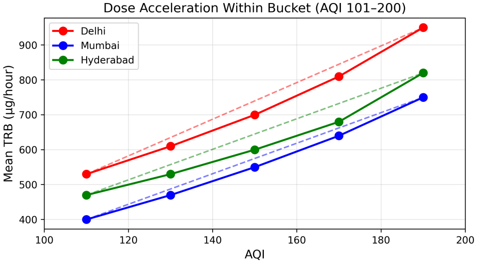
City	Increase in TRB when AQI goes 101 $\rightarrow$ 200
Delhi	+ 297 μg/hour
Mumbai	+ 370 μg/hour
Hyderabad	+ 305 μg/hour

- The same “Moderate” health advisory covers a three-fold range of actual pollutant mass entering your lungs.
- The gradient is steepest in Hyderabad (67% coefficient of variation).



Dose Accelerates at Higher AQI

- **Baseline slope (101–120):** 2.4–3.0 μg/h per AQI point.
- By **181–200**, slope is 2.3–2.6$\times$ steeper.
- Same 20-point AQI rise near “Poor” adds $>$2$\times$ more lung dose.



City	101–120	121–140	141–160	161–180	181–200
Delhi	3.0	1.2$\times$	1.6$\times$	2.0$\times$	2.3$\times$
Mumbai	2.4	1.3$\times$	1.8$\times$	2.2$\times$	2.4$\times$
Hyderabad	2.7	1.4$\times$	1.9$\times$	2.3$\times$	2.6$\times$

Example: Delhi’s 181–200 slope = 2.3 $\times$ 3.0 = 6.9 μg/h per AQI point.

Same AQI Number, Different Physiological Dose

- Same AQI formula everywhere, but dose varies by city – pollution mixtures differ.

TRB (μg/hour) at selected AQI (male sitting, mid-uptake):

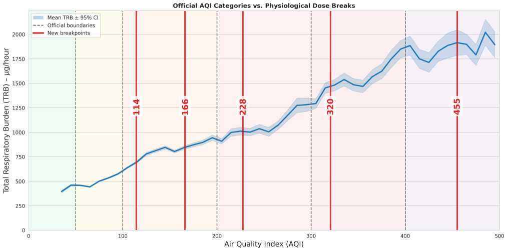
AQI	Delhi	Mumbai	Hyderabad	AQI	Delhi	Mumbai	Hyderabad
120	560	400	470	220	900	780	820
150	650	510	560	250	1020	900	940
180	740	620	660	280	1140	1020	1060

- **Observation:** Same AQI $\rightarrow$ different dose (e.g., AQI 180: Delhi > Hyderabad > Mumbai).
- **Why?** Hyderabad dominated by CO/NO_2 (high uptake); Mumbai by PM_{10} (lower per-μg dose).

Thresholds Discovery: Data-driven Breakpoints

- We computed optimal AQI breakpoints directly from TRB using two methods:
  - **Decision Tree** (forced 5 splits)
  - **GMM** on 7-pollutant space (6 clusters $\rightarrow$ 5 midpoints)
  - **Mean** of both methods

Method	**Breakpoints (AQI)**
DT	118, 212, 304, 380, 542
GMM	110, 121, 152, 262, 369
Mean (DT+GMM)	**114, 166, 228, 321, 455**
Official	**50, 100, 200, 300, 400**



The official colour-coded categories are systematically misaligned with where physiological dose actually changes.

Interpretation of Key Findings

- **Max-pooling kills 77–94%** of dose info – categories add little extra loss.
- **CO is the most informative** pollutant ($3.8\times$ PM10 in Hyderabad) – but drives AQI <8% of hours.
- **Same AQI, different dose across cities** – due to different pollutant mixtures (e.g., AQI 180: Delhi 740, Mumbai 620, Hyderabad 660 μg/h).
- **Synergy lost:** PM10 + CO together reveal more than sum – max-pooling discards it.
- **Inside "Moderate", dose triples** – and rise is non-linear (steepest near "Poor").

Limitations

- TRB is a dose metric, not a health outcome – but dose is causally upstream of health.
- Gas uptake fractions have 15–16% uncertainty – however, all ratios (compression, RMSE) are nearly invariant.
- Hospital data not used – would strengthen external validity, but TRB provides a clean, confounder-free audit.
- Severe category analysis is Delhi-only (Mumbai and Hyderabad record zero Severe hours).

Conclusion – Four Takeaways

1. **Max-pooling, not categories, is the real culprit.** (77–94% loss already at continuous AQI)
2. **AQI is blind to most informative pollutant.** CO carries 47–58% nats MI.
3. **Same AQI $\neq$ same dose.** Example: AQI 180 $\rightarrow$ Delhi 740, Mumbai 620, Hyderabad 660 μg/h. And "Moderate" hides a 3$\times$ dose range.
4. **Official breakpoints misaligned.** Data-driven: 114, 166, 228, 321, 455 vs. official 50, 100, 200, 300, 400.

Recommendations and Future Work

Recommendations (actionable):

- Replace max-pooling with a weighted additive index.
- Move away from 24-hour rolling averages.
- Adopt city-specific or regional thresholds.

Future work:

- Validate TRB against hospital admission data.
- Extend TRB to include VOCs and heavy metals.
- Build a real-time dashboard showing both AQI and TRB for public awareness.
- Transfer learning across cities – can a dose-prediction model trained in Delhi generalise?

Thank You

Questions?